Cubic Discrete Diffusion: Discrete Visual Generation on High-Dimensional Representation Tokens

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Research Questions

- How can discrete generative models handle high-dimensional representation tokens (768-1024 dimensions) instead of low-dimensional compressed tokens (8-32 dimensions)?
- Can dimension-wise quantization preserve semantic capabilities of high-dimensional features for both understanding and generation tasks?
- How can masked diffusion be extended to efficiently model the exponentially large space of h×w×d discrete tokens?

Core Idea

Current discrete visual generation methods are limited to low-dimensional tokens (8-32 dims), sacrificing semantic richness needed for understanding tasks. This creates a fragmented approach where vision uses high-dimensional features for understanding but low-dimensional tokens for generation. The key insight is to perform fine-grained masking across the entire h×w×d tensor space rather than masking entire spatial positions. Any dimension at any position can be independently masked and predicted, allowing the model to learn both intra-position correlations (how dimensions relate within a location) and inter-position patterns (spatial propagation) through bidirectional attention over partial observations.

Key Formula

Dimension-wise quantization

$$q_{x,y,i} = \text{Quantize}(z_{x,y,i}; L)$$

z_{x,y,i} is the i-th dimension at spatial position (x,y), L is number of quantization levels. Unlike vector quantization, this treats each dimension independently, making discretization tractable at 768+ dimensions.

Masking ratio sampling

$$r \sim \text{TruncNorm}(\mu = 1.0, \sigma, [0, 1.0])$$

r is masking ratio, μ=1.0 biases toward aggressive masking, σ controls diversity of masking scenarios. Distribution truncated to [0,1] for consistency with inference progression from fully masked to fully unmasked.

Key Figure

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Abstract

Visual generation with discrete tokens has gained significant attention as it enables a unified token prediction paradigm shared with language models, promising seamless multimodal architectures. However, current discrete generation methods remain limited to low-dimensional latent tokens (typically 8-32 dims), sacrificing the semantic richness essential for understanding. While high-dimensional pretrained representations (768-1024 dims) could bridge this gap, their discrete generation poses fundamental challenges. In this paper, we present Cubic Discrete Diffusion (CubiD), the first discrete generation model for high-dimensional representations. CubiD performs fine-grained masking throughout the high-dimensional discrete representation—any dimension at any position can be masked and predicted from partial observations. This enables the model to learn rich correlations both within and across spatial positions, with the number of generation steps fixed at T regardless of feature dimensionality, where $T \ll hwd$. On ImageNet-256, CubiD achieves state-of-the-art discrete generation with strong scaling behavior from 900M to 3.7B parameters. Crucially, we validate that these discretized tokens preserve original representation capabilities, demonstrating that the same discrete tokens can effectively serve both understanding and generation tasks. We hope this work will inspire future research toward unified multimodal architectures. Code is available at: https://github.com/YuqingWang1029/CubiD.

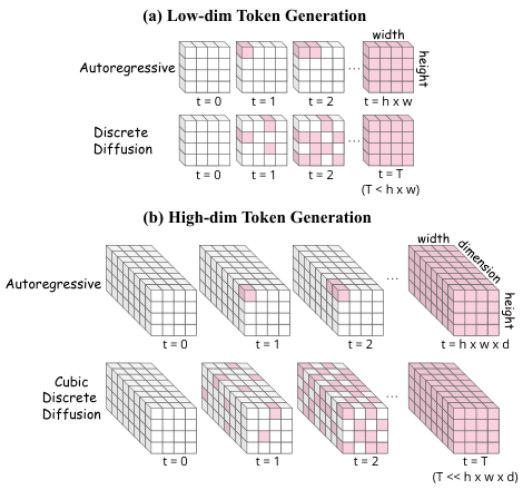

Figure 1. **Comparison of discrete visual generation approaches.** (a) **Low-dimensional token generation:** Both methods operate at the spatial level—autoregressive requires $h \times w$ sequential steps, while discrete diffusion achieves parallel generation in $T < h \times w$ iterations. (b) **High-dimensional token generation:** Autoregressive becomes intractable ($h \times w \times d$ steps), and standard discrete diffusion cannot model intra-position dependencies. Our Cubic Discrete Diffusion performs fine-grained masking across the entire 3D tensor—any dimension at any position can be masked and predicted independently—enabling efficient generation in $T \ll h \times w \times d$ iterations while capturing both spatial and dimensional correlations.

Figure 1: Illustrates the fundamental difference between low-dimensional and high-dimensional token generation, showing how CubiD enables fine-grained masking across the entire 3D tensor while traditional methods fail to scale.

Limitations

- Dependence on representation encoder quality - reconstruction PSNR ~18dB limits fine-grained details in generated images
- Gap with continuous generation methods like RAE despite discrete advantages for unified modeling
- Inference efficiency - requires hundreds to thousands of steps compared to continuous diffusion models
- Limited by dimension-wise quantization which may not capture cross-dimensional correlations as effectively as continuous representations